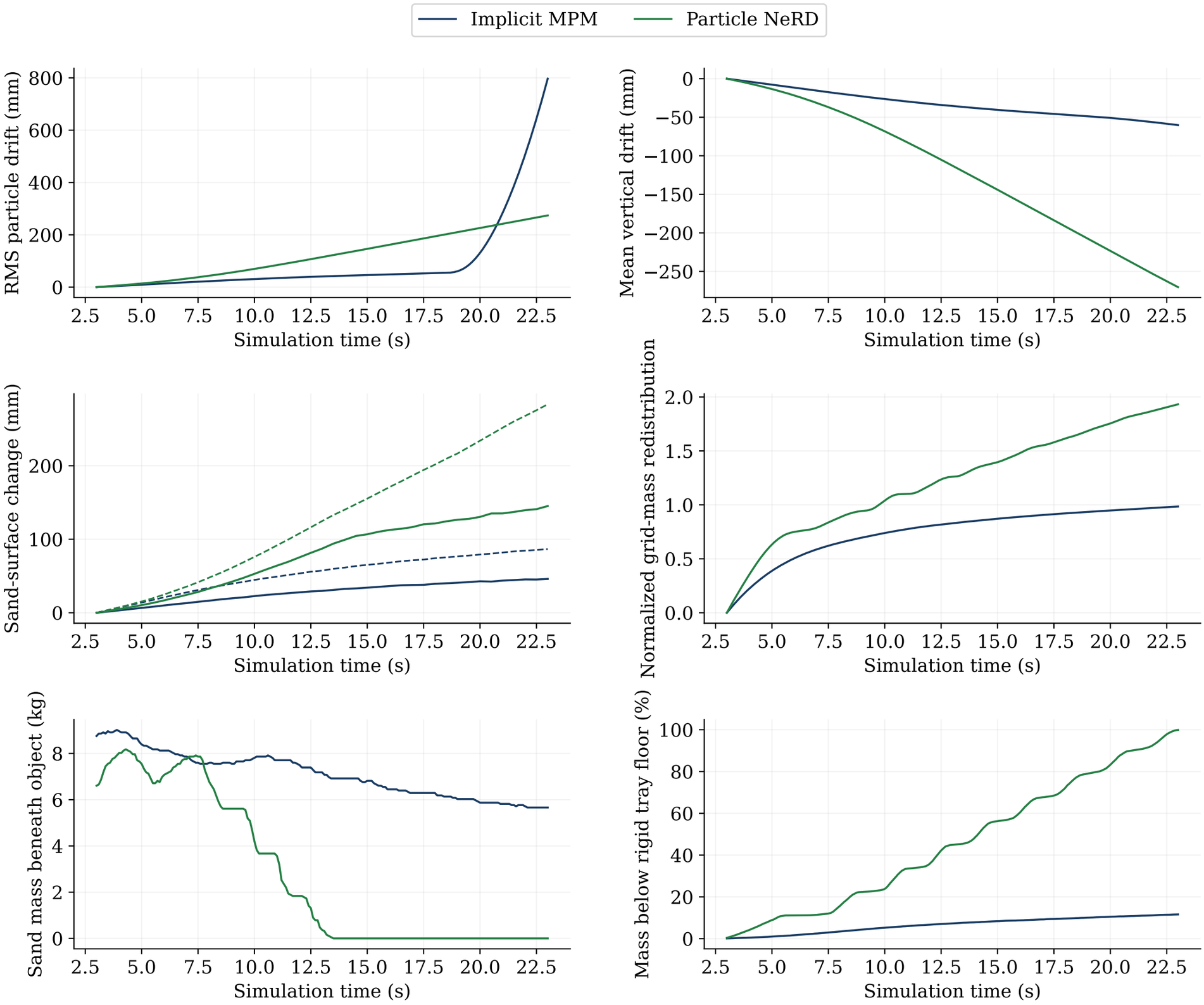

Figure 7: Fixed-object hold, measured at 3–23 s. Both methods start from identical particles at 0 s.
Top row: mass-weighted RMS drift from each method's own 3 s state; and mass-weighted mean vertical drift (negative down).
Middle row: distance to that method's 3 s material isosurface: solid mean, dashed 95th percentile (includes sides/bottom).
and sum of absolute grid-cell mass changes from 3 s, divided by total physical mass; dimensionless, range 0–2.
Bottom row: absolute mass in the rectangle enclosing the object's world x/y projection, extending 0.10 m below its
lowest z and clipped above the tray floor at z = −0.30 m. and absolute mass fraction with centers below that floor.
The object stops at 2 s. The 3 s anchor does not reset particles or assert equilibrium. One paired run per shape.
No alignment, gate, projection, filtering, or particle removal; MPM settling and leakage remain included.